YEAR 7 · UNIT 1 · HOMEWORK 2

# Signs and Animal Cells

Mathematics 1.2 — Multiplying & Dividing Integers
Science 1.2 — Animal Cells

**Name:** ____________________________ **Class:** ____________________________

**Date given:** _______________________ **Date due:** _________________________

### How to do this homework

- Write your answers **on this sheet**, in the spaces given. If you need more room, use the back of the page.
- In maths, **write the calculation before the answer**. Work out the **sign** first — it is half of every answer on this sheet.
- In science, answer in **full English sentences**. “Nucleus” is not a sentence; “The nucleus is the control centre of the cell” is.
- If a word is new to you, look for the blue **Word help** box on that page before you give up.

## Section A · Mathematics 1.2 — Multiplying and Dividing Integers

### Remember from the lesson

Dividing **undoes** multiplying, so **$\times$ and $\div$ obey the same four rules**:

| | | | |
|---|---|---|---|
| **Same** signs | the answer is **positive** | $-6 \times -7 = 42$ | $-42 \div -7 = 6$ |
| **Different** signs | the answer is **negative** | $-6 \times 7 = -42$ | $-42 \div 7 = -6$ |

The **product** of two numbers is the answer when you **multiply** them. The product of 2 and $-9$ is $-18$.

**Careful.** “Two negatives make a positive” is only true for **$\times$ and $\div$**. For **adding** it is false: $-3 + -4 = -7$, because adding a negative still sends you **left**.

### A1 Get the sign first

**(a)** Complete the four rules. Write **positive** or **negative** in each gap.

(i) positive $\times$ positive is ____________

(ii) positive $\times$ negative is ____________

(iii) negative $\times$ positive is ____________

(iv) negative $\div$ negative is ____________

**(b)** **Do not work out the digits.** Write only + or − for the sign of each answer.

(a) $-13 \times -8$ _____
(c) $-25 \times 16$ _____
(e) $99 \times -99$ _____
(b) $47 \div -1$ _____
(d) $-144 \div -12$ _____
(f) $-360 \div 9$ _____

## A2 Work them out

Now do the digits as well as the sign.

(a) $-7 \times 3 =$ _______
(e) $-10 \times 8 =$ _______
(i) $56 \div -7 =$ _______
(b) $-8 \times -4 =$ _______
(f) $0 \times -23 =$ _______
(j) $-72 \div -8 =$ _______
(c) $9 \times -6 =$ _______
(g) $-45 \div 5 =$ _______
(k) $-100 \div 4 =$ _______
(d) $-5 \times -11 =$ _______
(h) $-36 \div -6 =$ _______
(l) $27 \div -27 =$ _______

## A3 Say it, then write it

**Word help — these words decide the order of your calculation**

| | |
|---|---|
| **multiply −2 by 5** | $-2 \times 5$. |
| **the product of** | the answer you get when you **multiply**. The product of −3 and −2 is 6. |
| **divide −18 by 3** | $-18 \div 3$. The number being **shared out** is −18. |
| **3 divided into −18** | the **same** calculation, $-18 \div 3$. The English swaps the two numbers round; the calculation does not. |
| **share between** | divide it out equally. |
| **a debt** | money you must pay back. It makes a total **negative**. |
| **each / every / per** | one lot, repeated — and repeated adding is **multiplying**. So **falls by 4 every hour** means each hour the change is −4. |

Turn each English sentence into a calculation, then work out the answer. **Write the calculation first.**

1. Multiply −6 by 7.
   Calculation: ________________________________ Answer: ____________

2. Find the product of −4 and −9.
   Calculation: ________________________________ Answer: ____________

3. Divide −56 by 8.
   Calculation: ________________________________ Answer: ____________

4. Work out 5 divided into $-40$.
   Calculation: ______________________ Answer: __________

5. The temperature falls 4 degrees every hour for 7 hours. What is the total change?
   Calculation: ______________________ Answer: __________

6. A debt of 60 dollars is shared equally between 4 people. Write the amount **each** person owes as a negative number.
   Calculation: ______________________ Answer: __________

## A4 Brackets first

**Remember: the new rule does not live inside the bracket**

Work out **what is inside the bracket** first. Inside it you are often **adding**, and adding a negative still sends you **left**: $(-3 + -2) = -5$, **not** 5.

**(a)** Work these out. Show what is inside the bracket first.

(i) $(-3 + -5) \times 4 =$ ________

(ii) $(2 - 7) \times -6 =$ ________

(iii) $(-4 + -6) \div -5 =$ ________

(iv) $(-9 - -3) \div 2 =$ ________

(v) $-8 \times (-2 + 2) =$ ________

(vi) $(10 + -4) \times -3 =$ ________

**(b)** A student wrote $(-3 + -5) \times 4 = 32$.
In one full sentence, explain which rule they have used in the wrong place.

## A5 Catch it with an estimate

**Remember: an estimate is a rough answer you work out first**

**Estimate:** round each number to something easy, then calculate. If your real answer is nowhere near it, you have made a mistake. Example: $-4.1 \times 2.8$ rounds to $-4 \times 3 = -12$.

**(a)** Mr Bowen works out $-4.9 \times 21$ and gets 102.9. Estimate the answer by rounding both numbers. Estimate: ____________

**(b)** His answer must be wrong. Say what he has forgotten.

## A6 Word problems

Read each one **all the way to the end** before you start writing. Show the calculation, then the answer.

**1. The quiz game.** In Mr Bowen's quiz game, every wrong answer scores $-15$ points.

(a) A student gets 6 answers wrong. What is the change to their score?

(b) In a later game, a student's score changes by $-120$ points from wrong answers alone. How many did they get wrong?

**2. Eight forgotten debts.** Mr Bowen owes 8 different friends 7 dollars each.

(a) Write his total debt as a negative number.

(b) All 8 friends then tell him to forget the debt, so all 8 debts of $-7$ dollars are **taken away** from his total. Write this as a multiplication and work out the change.

(c) Is Mr Bowen better off or worse off than before? By how much?

**3. Mr Bowen sits his own quiz.** A correct answer scores 4 points and a wrong answer scores $-4$ points. Mr Bowen answers all 20 of his own questions. He gets 10 right and 10 wrong.

(a) How many points does he get from the correct answers?

(b) How many points does he get from the wrong answers?

(c) What is his total score?

## A7 Now you write the question

**Your turn to be the teacher**

So far the questions have given you the English, and you have written the maths. Now do it **backwards**. You are given the maths — write the English.

Your word problem must:

- be about something **real** — money, temperature or points in a game;
- use a word from the **Word help** box, such as **each**, **every**, **per**, **shared between** or **a debt**;
- be written in **full English sentences**, and end with a question;
- have the answer that is already given to you.

**(a)** Write a word problem for $5 \times -8 = -40$

**(b)** Write a word problem for $-60 \div 12 = -5$

## Section B · Science 1.2 — Animal Cells

**Remember from the lesson**

**You are an animal**. An animal cell has a **cell membrane**, **cytoplasm**, **mitochondria** and a **nucleus**. It has **no cell wall**, so it has **no fixed shape**. A plant cell has all of those **plus** a **cell wall**, **chloroplasts** and a **sap vacuole**.

## B1 The words for comparing two things

**Word help — how to compare in English**

| | |
|---|---|
| **similar** | alike in some ways, different in others. |
| **the same** | no differences at all. It is **stronger** than **similar**. |
| **Both A and B …** | the thing that follows is true of **each** of them. |
| **…, but …** | joins two halves that **disagree**. |
| **Unlike A, B …** | B is **different** from A in the way that follows. |

Fill each gap with one of: **Both**, **and**, **but**, **Unlike**, **have**, **similar**, **the same**. Each word is used once.

1. ___________ a plant cell ___________ an animal cell have a nucleus.

2. A plant cell has chloroplasts, ___________ an animal cell does not.

3. ___________ a plant cell, an animal cell has no fixed shape.

4. Both kinds of cell ___________ mitochondria.

5. An animal cell is ___________ to a plant cell, but the two are not ___________.

## B2 True or false?

Tick one box for each sentence. Then **rewrite each false sentence** on the lines below so that it becomes true.

| **Sentence** | **True** | **False** |
|---|---|---|
| 1. Every animal cell has a nucleus. | | |
| 2. An animal cell has a cell wall made of cellulose. | | |
| 3. Mitochondria are found in plant cells and in animal cells. | | |
| 4. An animal cell has no fixed shape because it has no cell wall. | | |
| 5. A plant cell has three parts an animal cell has not. | | |
| 6. Chloroplasts are green because they contain cellulose. | | |

Rewrite the false sentences here, in full:

## B3 Answer in full sentences

**Sentence starters — you may copy these**

"Every animal cell has..."
"An animal cell has no fixed shape because..."
"Both a plant cell and an animal cell have..., but only..."
"Unlike a plant cell, an animal cell..."

1. Name the **four** parts that every animal cell has.

2. A plant stands still, but an animal moves — it bends, squeezes and changes shape all day. Explain why an animal cell has **no fixed shape**. Use the word **because**.

3. Give **two** ways an animal cell is different from a plant cell. Write each one as a full sentence, using **but** or **Unlike**.

## B4 Two pictures down a microscope

Mr Bowen looks at two slides. **Picture 1** shows cells packed tightly together like bricks in a wall, each one with straight, stiff edges. **Picture 2** shows soft, rounded cells with no straight edges anywhere.

**(a)** For each picture, say whether it shows **plant** cells or **animal** cells, and say **how you know**. Use the word **because**.

Picture 1:

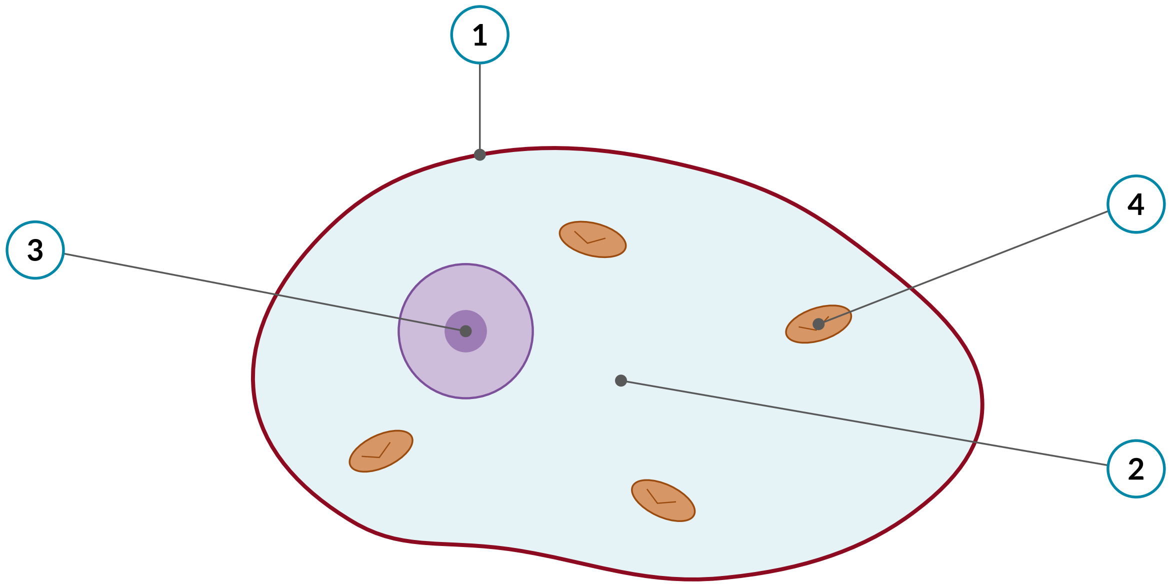
Picture 2:

**(b)** Neither picture has anything green in it. A student says: *"There is no green, so both pictures must show animal cells."*

Explain in one full sentence why the student is wrong. (Hint: think about where an onion grows.)

## B5 Label the animal cell

Write the name of each numbered part on the lines below the diagram.




1. ______________________ 3. ______________________

2. ______________________ 4. ______________________

**(b)** Name the **three** parts you would have to **add** to this diagram to turn it into a **plant** cell.

______________ ______________ ______________

## Answer Key — Teacher Copy

### A1 Get the sign first

(a) (i) positive (ii) negative (iii) negative (iv) positive.
(b) (a) + (b) − (c) − (d) + (e) − (f) −

This is the easy question on purpose — everybody should finish it. If a student is doing the arithmetic in (b), stop them: the instruction says not to. Getting the sign without the digits is the skill being tested.

### A2 Work them out

(a) −21 (b) 32 (c) −54 (d) 55 (e) −80 (f) 0
(g) −9 (h) 6 (i) −8 (j) 9 (k) −25 (l) −1

Watch (f): a few will write −0 or leave it blank because the rule does not seem to cover it. Zero has no sign, and anything times zero is zero.

### A3 Say it, then write it

1. $-6 \times 7 = -42$
2. $-4 \times -9 = 36$
3. $-56 \div 8 = -7$
4. $-40 \div 5 = -8$
5. $7 \times -4 = -28$
6. $-60 \div 4 = -15$

Question 4 is the one that matters. "5 divided into −40" puts the 5 first in the English and second in the calculation. A student who writes $5 \div -40$ has made the *English* error, not an arithmetic one — make them read the sentence again and ask which number is being shared out. It is the same trap as "subtract 5 from 8" in Homework 1, and it will come back every term.

Question 6: accept −15 or "each person owes 15 dollars", but the question asks for a negative number, so push for −15.

### A4 Brackets first

(a) (i) −32 (ii) 30 (iii) 2 (iv) −3 (v) 0 (vi) −18

(b) They have used the × and ÷ rule inside a bracket that contains an **addition**. Two negatives only make a positive when you multiply or divide, so $-3 + -5 = -8$, not 8, and the answer is −32.

**This is the question of the sheet.** Expect a large part of the class to get (b) wrong, or to write "they added instead of multiplying". That is worth ten minutes of whole-class discussion rather than a red pen: the misconception is the sentence itself, not the sum. Ask them to say out loud which *operation* the rule belongs to.

## A5 Catch it with an estimate

(a) $-5 \times 20 = -100$, so the answer should be about $-100$.

(b) His answer is *positive*, so he has forgotten the sign: a negative times a positive is negative. The digits were fine — the true answer is $-102.9$.

Accept any sensible rounding: $-5 \times 21 = -105$ is just as good. Do not mark the estimate against the exact answer — mark whether it was close enough to catch the error. That is the only reason estimating is on this sheet, and it is why the four rounding drills that would normally sit here are not: Workbook Exercise 1.2 Q8 and Q9 already set exactly those, so repeating them here would waste the homework.

## A6 Word problems

1. (a) $6 \times -15 = -90$ points. (b) $-120 \div -15 = 8$ questions.
   Part (b) is the only place on this sheet where a negative divided by a negative appears in a real context. The answer is a *count*, so it has to come out positive — that is a good sanity check to point out.

2. (a) $8 \times -7 = -56$ dollars. (b) $-8 \times -7 = +56$, so his total changes by +56 dollars. (c) Better off, by 56 dollars — he is back to zero.
   This is the honest real-world meaning of "negative times negative": taking away 8 debts. If a student writes $8 \times -7$ for (b) and gets $-56$ again, ask them whether cancelling a debt makes you richer or poorer. Most will answer that correctly out loud and then see their own sign error.

3. (a) $10 \times 4 = 40$. (b) $10 \times -4 = -40$. (c) $40 + -40 = 0$.
   The deadpan one. Mr Bowen has scored zero on his own quiz and is very pleased with himself. Part (c) is an *addition*, so this is one more place the false version of "two negatives make a positive" can be caught.

## A7 Now you write the question

There is no single right answer. Check the four conditions in the green box, in this order:

1. **Is it a repeated lot, or a sharing out?** (a) must be *five lots of* $-8$ — five hours, five people, five wrong answers. (b) must be a total of $-60$ *split* into 12 equal parts. Turning (b) into a multiplication is the most common miss.

2. **Does the signal word match?** (a) needs **each**, **every** or **per**. (b) needs **shared between**, **divided by** or an equal rate.

3. **Full sentences, ending in a question?**

4. **Does their own answer come out as $-40$, and as $-5$?**

Two that work: (a) "Mr Bowen loses 8 dollars every day for 5 days. What is the change to his money?"
(b) "A diver goes down 60 metres in 12 minutes at a steady rate. What is the change in her depth each minute?"
Expect (b) to be harder than (a) by a distance, because a division story needs a *rate* and most of them will write a subtraction instead. Discuss it as a class before marking it.

## B1 The words for comparing two things

1. Both … and 2. but 3. Unlike 4. have 5. similar … the same

Sentence 4 catches the students who write "has". "Both kinds of cell" is plural, so the verb is *have*. Worth saying once, clearly.

## B2 True or false?

1. True 2. **False** 3. True 4. True 5. True 6. **False**

Rewrites: 2. "An animal cell has *no* cell wall." (Or: "A *plant* cell has a cell wall made of cellulose.") 6. "Chloroplasts are green because they contain *chlorophyll*."

Sentence 6 is a vocabulary trap, not a biology one: *cellulose* and *chlorophyll* both begin with c-h/c-e and both belong to plants. If the class mixes them up, the fix is that cellulose is the *building material of the wall* and chlorophyll is the *green colour that catches sunlight*.

## B3 Answer in full sentences

1. Every animal cell has a cell membrane, cytoplasm, mitochondria and a nucleus.
2. An animal cell has no fixed shape because it has no cell wall to hold it stiff, and an animal has to bend and move.
3. Any two, each a full sentence. For example: "An animal cell has no cell wall, but a plant cell has one." "Unlike a plant cell, an animal cell has no chloroplasts, so it cannot make its own food." "A plant cell has a large sap vacuole, but an animal cell does not."

Mark the *sentence*, not just the fact. A list of three words is not an answer to any of these, and saying so consistently is the only thing that changes it.

## B4 Two pictures down a microscope

(a) Picture 1 shows plant cells, because the cells have straight, stiff edges and are packed together like bricks — that is a cell wall, and only plants have one. Picture 2 shows animal cells, because the cells are soft and rounded with no straight edges, which means there is no cell wall.

(b) The student is wrong because not every plant cell has chloroplasts: an onion bulb grows underground in the dark, so its cells are not green, and they are still plant cells. Green is a clue that can fail; the *cell wall* is the clue that never does.

(b) is the hardest question in Section B and the one most worth discussing. "No green means animal" is the misconception the class arrived at in the lesson, and it needs breaking twice before it goes.

## B5 Label the animal cell

1 — cell membrane 2 — cytoplasm 3 — nucleus 4 — mitochondrion

(b) A cell wall, chloroplasts and a sap vacuole. Everything already on the diagram stays exactly where it is.

Accept "mitochondria" for 4. Do not accept "cell wall" for 1: this cell has no wall, and the leader points at the thin outline, which is the membrane. That confusion is the reason the outline here is drawn as a single thin line with no green band around it.